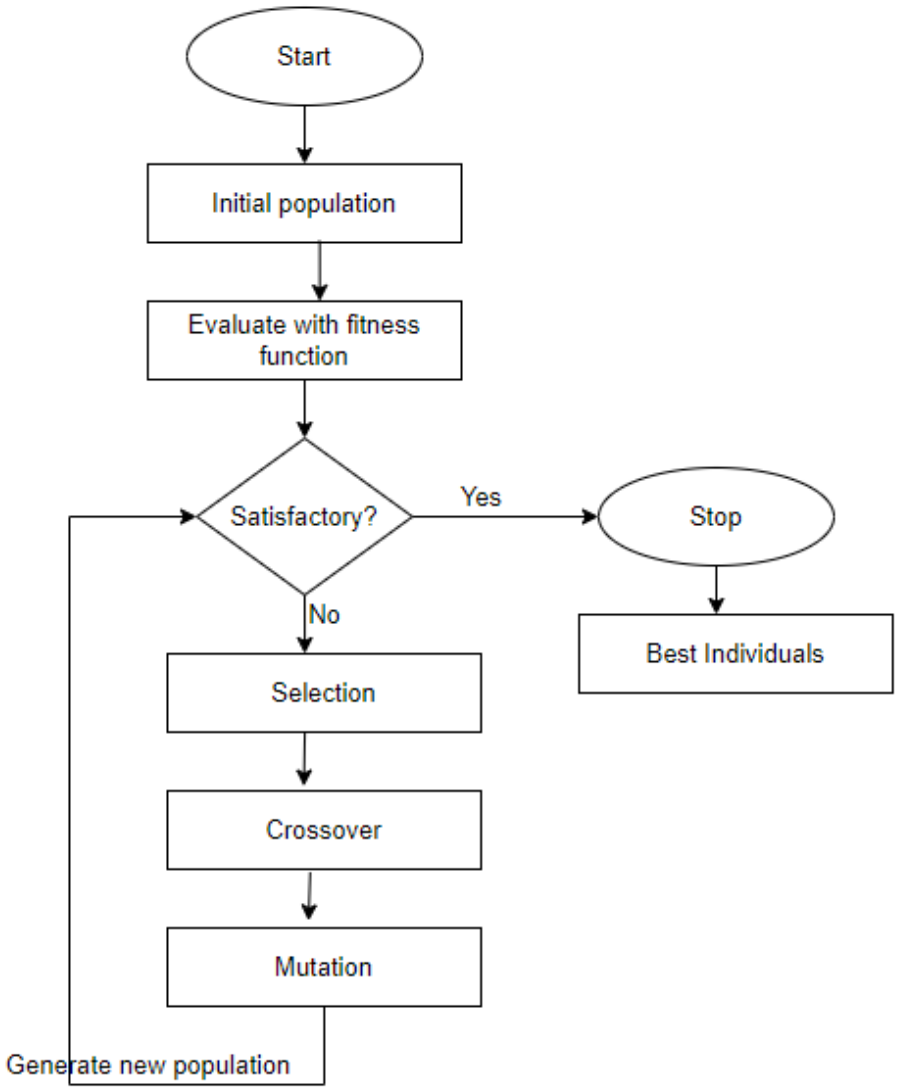
Project 2

Implement a genetic algorithm to generate the string

Hello CPSC 390!

starting from a population of randomly-generated strings (characters from ‘ ‘ to ‘z’ in ASCII).

Use a population size of 128; you will choose how to implement the fitness evaluation, selection, crossover, and mutation procedures



Submission:

Please use Jupyter Notebook for the implementation. Your submission should include code, output, and documentation, all in one notebook file. Clearly document your representation of the individuals and how you implement the fitness evaluation, as well as selection, crossover, and mutation processes using notebook text cells. All three processes must be stochastic in nature, and the initial population must be randomly generated.

Output:

1. Print the highest-fitness string of every 10th generation
2. plot the fitness score of the best individual in every generation using a log scale for the horizontal axis

Discussion: analyze how effectively (or ineffectively) your genetic algorithm converges on the target string. Explain why and discuss how you can improve your algorithm without increasing the population size. (Ref. Section 4.1.4)

Each student should complete the project individually. You can use utility packages, but you must implement the core algorithm.

Grading:

- 70% for successful execution and correct output
- 10% for methodology documentation, code structure, and readability
- 20% for discussion on the effectiveness of your algorithm
- Extra credit: up to 20% if you implement the improvement idea(s) and demonstrate the improvements by measuring its performance over 32 iterations vs. 32 iterations of the original method